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Deduce that L is tilted at 32° to the vertical, so

$$L \sin 32^\circ = \frac{mv^2}{r} \quad \text{--- (1)}$$

$$\text{and } L \cos 32^\circ = mg \quad \text{--- (2)}$$

$$(1) \div (2) \text{ gives: } \tan 32^\circ = \frac{v^2}{rg}$$

When r is halved to $r/2$ and v is doubled to $2v$, let the angle be ϕ .

$$\text{We now have: } \tan \phi = \frac{(2v)^2}{\left(\frac{r}{2}\right)g} = 8 \left(\frac{v^2}{rg} \right) = 8 (\tan 32^\circ)$$

$$\therefore \phi = 78.7^\circ$$